

Inuka Hackathon

1. Setting of Each Stage

All hackathon days follow a standard full-day cadence at **Em-Tech**, with mentors and Em-Tech supervisors acting as "clients" on the floor. Teams retain the same challenge domain across all stages.

- **Stage 1: Data Engineering Hackathon**
 - **Date:** Friday, 24 July 2026
 - **Setting:** Full-day, heads-down build sprint at Em-Tech, focusing on foundational data pipelines.
- **Stage 2: Operational Analytics Hackathon**
 - **Date:** Friday, 21 August 2026
 - **Setting:** Full-day build sprint at Em-Tech, focusing on analytics, diagnostics, and prediction.
- **Stage 3: Solutions Showcase**
 - **Date:** Wednesday, 9 September 2026
 - **Setting:** Showcase day (not a build day; code freeze is in effect) at Em-Tech. Features rehearsed demonstrations, extended Q&A, and ROI pitches to judges (Em-Tech senior staff, PLP, and KPC representatives).

2. The 10 Problem Statements

Domain A: Pipeline Integrity & Product Loss

1. **Shrinkage Detection:** Detect and quantify real-time product losses and anomalies along the pipeline network using flow, meter, and pressure data.
2. **Unaccounted-for Product Prediction:** Predict future product shrinkage to proactively reduce unaccounted-for product before it impacts the supply chain.

Domain B: Predictive Maintenance & Equipment Reliability

3. **Condition-Based Maintenance Transition:** Anticipate pump, valve, and sensor failures using maintenance logs and telemetry to shift KPC from reactive to condition-based maintenance.
4. **4. Downtime Reduction:** Cut operational downtime by integrating tested schedulers/APIs into existing maintenance workflows with performance monitoring.

Domain C: Depot, Terminal & Supply Chain Operations

5. **Throughput & Turnaround Optimization:** Improve depot throughput and truck-loading turnaround times to directly reduce queuing costs and demurrage.
6. **6. Regional Demand & Stock-out Prevention:** Optimize product movement and forecasting to match regional demand, preventing costly stock-outs.

Domain D: Revenue Assurance: Billing & Reconciliation (Fintech)

7. **Order-to-Cash Leakage Closure:** Reconcile product loading and dispatch events against invoicing and payments to close revenue-leakage gaps in KPC's order-to-cash cycle.
8. **E-Billing Integration & Audit:** Prototype e-billing integrations and build an exceptions dashboard with a clear audit trail for financial transparency.

Domain E: HSE, Compliance & Deployment Readiness

9. **Proactive HSE Early-Warning:** Turn retrospective safety, environmental, and compliance records into proactive early-warning analytics and automated alert narratives.
10. **Secure, Deployable AI Tooling:** Ensure messy operational data is ingested, cleaned, and anonymized (to meet KPC data sensitivity rules) and shipped as a hardened, monitored CI/CD service with an executive-facing view.

3. Grading Rubric

A **single cumulative rubric** applies across all three stages. The final "Top 3" is determined by the total series score, ensuring consistency and rewarding sustained effort rather than a single day's performance.

Stage Weighting (Cumulative):

- Stage 1: 20%
- Stage 2: 35%
- Stage 3: 45%

Evaluation Criteria:

- **Relevance to the KPC problem (25%):** The solution addresses a named, validated KPC pain point; scope has not drifted.
- **Technical quality (25%):** Sound engineering: reliable pipelines, defensible models, reproducible work, QA evidence (test coverage, data-quality gates), and CI hygiene.
- **Insight, impact & ROI (20%):** Quantified operational benefit (e.g., liters of loss avoided, downtime hours saved); honest assumptions; credible cost-benefit narrative.
- **Usability & deployability (15%):** Realistic adoption potential for Em-Tech or KPC; deployment pipeline, monitoring/support plan, documentation, and handover readiness.
- **Presentation & storytelling (15%):** Clear, confident communication pitched at an operational leadership audience.

4. Learner Expectations for Each Stage

Stage 1: Data Engineering (20% of total score)

- **Focus:** "Build the pipeline for the pipeline."
- **Learner Expectations & Deliverables:**
 - Ingest messy operational domain data and deliver a clean, documented, automated ETL foundation.
 - Establish a version-controlled repository with basic CI and data-quality checks/gates from day one.
 - Draft a one-page problem-framing memo linking the data to the specific KPC problem.

- **Presentation:** Deliver a tight 5-minute pitch focusing on the KPC problem, solution, evidence, and impact.

Stage 2: Operational Analytics (35% of total score)

- **Focus:** Build intelligence and prediction on top of the Stage 1 foundation.
- **Learner Expectations & Deliverables:**
 - Develop an analytics package: dashboards, statistical diagnostics, and first predictive models.
 - Quantify the KPC impact with specific, honest metrics (e.g., downtime hours saved, demurrage costs reduced).
 - Deliver the work as a QA'd beta complete with test evidence and User Acceptance Testing (UAT) notes.
 - **Presentation:** Deliver a 10-minute presentation + Q&A per team.

Stage 3: Solutions Showcase (45% of total score)

- **Focus:** Hardening, deployment, and business case (**No new scope**; code freeze is in effect).
- **Learner Expectations & Deliverables:**
 - Harden and ship a deployable, AI-powered tool via a CI/CD pipeline to a running, monitored service with alerting.
 - Provide comprehensive documentation, a support plan, and an executive-facing view of the tool.
 - Draft and present a quantified Return-on-Investment (ROI) pitch/case with supervisor input.
 - **Presentation:** Deliver a 10-minute demo and ROI pitch + Q&A per team.